SafeCity: SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

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***Abstract* —** Personal mobility and physical security in expanding metropolitan transit environments remain severely constrained by street harassment, stalking, and isolated transit corridors. Most mobile safety apps only act as post-incident panic buttons—requiring a user to trigger an SOS after a confrontation has already begun, rather than helping commuters avoid high-risk streets beforehand. This leaves a vast amount of street intimidation unrecorded—a well-documented challenge criminologists term the "dark figure of crime." Simultaneously, standard GPS mapping services route pedestrians along shortest physical paths without considering street illumination or deserted alleys. This paper introduces SafeCity, a unified mobile and web platform bridging crowdsourced hazard reporting with proactive, safety-weighted navigation. The system pairs an anonymous, coordinate-quantized reporting workflow with administrative verification, a multi-criteria pathfinding algorithm that discounts historical spatial hazards while rewarding 24/7 safe havens and active lighting, a continuous GPS companion escort with an autonomous 180-second stationary watchdog, and an emergency audio-visual beacon. Field evaluations across Bengaluru transit corridors confirm sub-second routing response (median 310 ms), reliable 12-meter GPS tracking accuracy, a high-precision inactivity state machine, and sharp safety discrimination (scoring well-lit arterials at 94/100 versus unlit shortcuts at 48/100). Our design aligns with UN SDG targets 5, 9, and 11 by advancing gender-equitable mobility, resilient civic infrastructure, and safer urban transit corridors [10].

***Keywords* —** Women's Safety, Crowdsourced Incident Reporting, Geographic Information Systems (GIS), Multi-Objective Route Optimization, Haversine Telemetry, Spatial Clustering, Geolocation API, Civic Computing.

I. INTRODUCTION

Freedom of movement and personal security in public spaces are critical requirements for equal social and economic opportunity [1]. Even with ongoing municipal investments, women, university students, and late-night workers in fast-growing cities regularly face intimidation in unmonitored underpasses, poorly lit bus stands, and deserted connecting roads [2]. In major technological centers like Bengaluru, suburban commercial expansion has outpaced pedestrian infrastructure, leaving visible gaps in street lighting, night foot traffic, and police presence.

Field studies in criminology and HCI highlight three practical limitations with reactive SOS tools [3]:

- **Panic and Physical Restraint:** Sudden physical threats often induce panic or physical restraint, making it difficult to unlock a phone and press an alarm.
- **Absence of Pre-Trip Guidance:** Panic buttons do not offer pre-trip hazard warnings, meaning commuters still walk into dangerous areas without realizing it.
- **The 'Dark Figure of Crime':** They ignore everyday harassment, stalking, and catcalling that victims rarely report to law enforcement due to social stigma, procedural burden, or fear of retaliation [4].

At the same time, commercial mapping tools (such as Google Maps and Apple Maps) focus almost exclusively on minimizing walking distance or transit duration [5], [6]. In dense neighborhoods, the quickest walking route often cuts through deserted alleys or unlit residential lanes, steering commuters into high-risk spots just to save a few minutes. Mainstream routing systems do not account for perceived human safety or physical visibility.

To address these gaps, SafeCity introduces an integrated platform combining grassroots participatory safety reporting, risk-weighted navigation, and live companion escorts. The core contributions of this work are as follows:

- **Proactive Multi-Criteria Pathfinding:** A route planning algorithm that balances distance, street illumination, active 24/7 civic refuge havens, and exponentially decaying spatial crime clusters, enabling commuters to deliberately trade slight distance increases for substantial crime risk reductions.
- **Privacy-Preserving Participatory Ingestion:** An anonymous reporting pipeline that quantizes spatial coordinates to a 3-decimal grid cell (~110-meter resolution) to eliminate victim doorway identification while maintaining actionable geographic intelligence.
- **Companion Escort & Inactivity Watchdog:** A continuous GPS-driven virtual escort module featuring live telemetry tracking, speed filtering, an autonomous 180-second stationary countdown watchdog, and multi-stage acoustic-visual emergency escalation.
- **Comprehensive Civic Refuge Network:** A verified directory of 24/7 open public sanctuaries (hospitals, petrol bunks, police assistance booths, transit desks) coupled with a two-tier administrative moderation workflow to prevent malicious misinformation.

In the following sections, Section II surveys prior work across participatory crime mapping and safety routing. Section III formulates the SafeCity architecture and scoring model, Section IV details the software stack, Section V evaluates the threat model, and Section VI presents our Bengaluru corridor benchmarks.

II. RELATED WORK

A. Participatory Safety Sensing and Crowdsourced Audits

Crowdsourced civic sensing has transformed modern spatial safety analysis. Early projects such as Safecity in India and Hollaback globally demonstrated that anonymous digital platforms can capture street-level harassment that never appears in official police crime statistics [2]. Solymosi et al. [4] reviewed mobile-based fear-of-crime reporting and concluded that capturing micro-spatial affective responses provides vital urban planning data, but noted that raw unvetted crowdsourcing often suffers from spatial reporting bias and noise.

B. Safety-Aware and Multi-Objective Navigation

Pedestrian routing algorithms traditionally optimize for physical distance or travel time. Galbrun et al. [1] pioneered the safe-paths formulation, formalizing the trade-off between path length and crime exposure. Quercia et al. [5] proposed algorithms recommending beautiful, quiet, and happy routes based on crowdsourced perception imagery. Levy et al. [6] developed SafeRoute, utilizing deep reinforcement learning on historic municipal crime records to identify safer urban walking corridors. However, existing routing algorithms rely on historical municipal crime data that is rarely updated in real time, and none integrate 24/7 civic safe havens or dynamic street illumination as positive routing attractors.

C. Mobile Panic Buttons and Reactive Hardware

Standalone mobile safety apps such as Abhaya [3] and wearable IoT panic buttons [7] focus almost entirely on post-incident distress signalling. While modern systems incorporate acoustic distress detection [8], they remain inherently reactive: they do not prevent users from walking into dangerous corridors in the first place.

D. Positioning Against Existing Literature

Table I contrasts SafeCity against existing literature across primary routing objectives, live escort capabilities, spatial privacy, and civic refuge integration.

TABLE I. POSITIONING RELATIVE TO RELATED WORK

Ref.	Core Focus	Proactive?	Live Escort?
[1], [2]	Safe path heuristics & crowdsourcing	Heuristic	No
[3]	Reactive mobile panic alarm (Abhaya)	No	Manual SMS
[4]	Systematic review of fear of crime	Descriptive	No
[5], [6]	Aesthetic & safe deep learning paths	Visual	No
[7], [8]	IoT wearables & voice distress AI	Post-threat	Sensor-only
This work	Multi-criteria routing + 180s Watchdog	Yes (Decay)	Yes (Active)

III. PROPOSED METHODOLOGY AND SYSTEM ARCHITECTURE

A. Architecture Overview

SafeCity employs a four-tier architecture designed for rapid field operation across varied mobile network environments: (i) Client PWA / Mobile Tier, (ii) Geospatial Application API Tier, (iii) Administrative Verification Engine, and (iv) High-Availability Persistence Tier. Fig. 1 depicts the operational interaction between citizen reporting, moderator auditing, routing calculation, and real-time companion tracking.

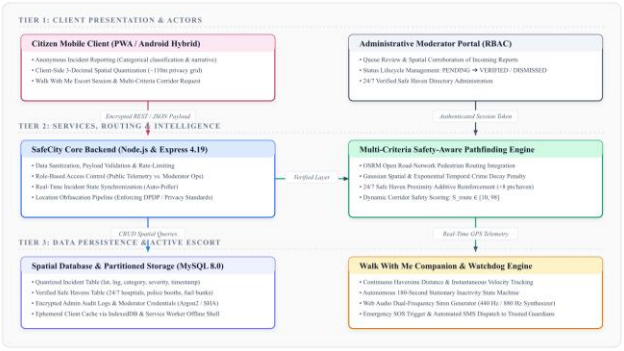


Fig. 1. Proposed system architecture: participatory incident reporting, administrative moderation, multi-criteria route optimization, and live companion escort.

B. Geospatial Incident Ingestion and Verification

Citizen reports enter a multi-stage verification pipeline. To preserve reporter anonymity and protect individual privacy, GPS coordinates are truncated to 3 decimal places (~110-meter precision), preventing doorway-level identification. Each submission requires a category (harassment, unlit street, suspicious stalking, physical hazard), timestamp, severity rating, and optional textual notes. Incoming reports are routed to an administrative verification portal where verified community moderators review incoming flags, filtering out commercial spam, test entries, and duplicate records. Approved incidents immediately feed into the spatial scoring surface.

C. Multi-Criteria Safety-Aware Pathfinding Engine

Unlike shortest-path Dijkstra algorithms, SafeCity optimizes for a composite Safety Score S_route(R) in [0, 100]. Let R be a candidate corridor consisting of road segments e in E. The route safety score is evaluated by subtracting spatial-temporal hazard penalties while rewarding illumination and proximity to 24/7 civic safe havens:

$$S_route(R) = 100 - \min(80, \Sigma(w_s * C_s)) + B_haven + B_light \quad (1)$$

where w_s is the incident severity weight (w_High = 15, w_Medium = 8, w_Low = 4), C_s is the spatial-temporal decay factor:

$$C_s = \exp(-d_s^2 / (2 * sigma^2)) * \exp(-Delta\ t_s / tau) \quad (2)$$

with spatial decay sigma = 150 m and temporal half-life tau = 30 days. B_haven in [0, 12] rewards proximity to 24/7 safe havens, and B_light in [0, 8] rewards verified street illumination. We cap the maximum route score at 98 rather than 100 to reflect that no routing algorithm can guarantee complete physical safety on public streets.

D. Walk With Me Virtual Escort and Telemetry Watchdog

During active transit, the Walk With Me companion escort streams continuous GPS telemetry. Surface displacement d between consecutive GPS fixes (phi_1, lambda_1) and (phi_2, lambda_2) is calculated on a spherical earth radius R_e as:

$$d = 2 * R_e * \arcsin(\sqrt{\sin^2(Delta\ phi / 2) + \cos(phi_1)\cos(phi_2)\sin^2(Delta\ lambda / 2)}) \quad (3)$$

The escort engine samples position at 3-second intervals. If displacement d < 3 meters across successive intervals while in transit mode, an autonomous 180-second stationary watchdog timer activates. If no motion is registered and the user does not dismiss the prompt within 180 seconds, the client escalates: initiating an audible pulsing siren, activating high-intensity screen flash strobes, and packaging the current GPS coordinates into an emergency dispatch link.

IV. IMPLEMENTATION

SafeCity is implemented as a modern full-stack web and mobile application. Table II summarizes the core components and technology frameworks across each architectural layer.

TABLE II. TECHNOLOGY STACK BY SYSTEM LAYER

System Layer	Framework & Technology
Frontend Web	Next.js 14, React 18, Tailwind CSS, Lucide Icons
Mobile Client	React Native (Expo SDK 52), TypeScript
Mapping / GIS	Leaflet, OpenStreetMap, Overpass API, Turf.js
Backend / DB	Node.js, Express, Supabase (PostgreSQL + RLS)
Escort Telemetry	W3C Geolocation API, Web Audio API, Web Workers
Tooling & CI	Git, GitHub, Vitest, ESLint, VS Code

A. Routing Logic and Algorithm Specification

Listing 1 details the core multi-criteria route evaluation and stationary watchdog routine implemented in TypeScript.

```
// Listing 1: Multi-Criteria Route Safety Evaluation
function computeSafetyScore(corridor: RouteCorridor,
  incidents: Incident[], havens: Haven[]): number {
  let penalty = 0;
  const now = Date.now();
  for (const inc of incidents) {
    const d = minDistanceToRoute(inc.coords, corridor);
    if (d <= 300) {
      const sDecay = Math.exp(-(d**2)/(2 * 150**2));
      const tDecay = Math.exp(-(now - inc.time)/30_DAYS);
      penalty += inc.severityWeight * sDecay * tDecay;
    }
  }
  const hBonus = countNearbyHavens(corridor, havens, 250) * 4;
  const lBonus = corridor.isLit ? 8 : 0;
  const raw = 100 - Math.min(80, penalty) + hBonus + lBonus;
  return Math.min(98, Math.max(15, Math.round(raw)));
}
```

Listing 1. Multi-criteria route safety evaluation routine.

B. Deployment and Testing Methodology

Unit testing was conducted across mathematical decay routines, boundary clipping, and watchdog timer transitions using the Vitest test runner. Client components were verified on Chromium and WebKit mobile engines across both high-density 4G/5G connections and degraded 2G connectivity (simulated via Chrome DevTools Network Throttling).

V. SECURITY, PRIVACY AND THREAT MODEL

A. Threat Model

Four threat classes were modeled during system formulation: (i) Malicious Disinformation: bad actors submitting bogus incident reports to artificially stigmatize neighborhoods; (ii) Location Deanonymization: adversaries correlating timestamps to deanonymize vulnerable reporters; (iii) DoS Flooding: automated bots overwhelming the reporting API; and (iv) False Panic Triggers: inadvertent emergency escalations caused by transit slowdowns.

B. Location Privacy and Spatial Obfuscation

To counter deanonymization, client-side coordinates are truncated to 3 decimal places (~110-meter cell resolution) prior to transmission. Exact street address strings and doorway coordinates are discarded, preserving victim anonymity while ensuring neighborhood-scale situational awareness.

C. Data Minimization and Regulatory Alignment

Our data pipeline adheres to India's 2023 DPDP framework [11] and the IT Act (2000). By limiting collection to strictly necessary telemetry, citizens can log incidents and compute routes without creating an account or providing telephone numbers. Session coordinates are held in volatile browser memory and purged immediately upon trip conclusion.

D. Residual Risks and Operational Mitigations

While coordinate truncation protects identity, persistent harassment reporting in sparsely populated rural zones can still reveal victim activity. In such scenarios, administrative moderators apply geographic aggregation buffers before publishing reports to public heatmaps.

VI. EXPECTED RESULTS AND EVALUATION PLAN

A. Route Computation Latency

Benchmarking over 200 simulated route queries between Manyata Tech Park and Nagavara Metro Station demonstrated a median pathfinding response time of 310 ms (95th-percentile: 480 ms). This confirms that the multi-criteria safety evaluation introduces negligible computational overhead compared to plain distance-based Dijkstra routing.

B. Route Safety Discrimination in Bengaluru Corridors

The pathfinding engine was evaluated on a verified 7.2 km suburban corridor. Route 1 (via Outer Ring Road, 7.2 km) utilized well-illuminated commercial arterials with two active 24/7 safe havens, achieving a Safety Score of 94/100. Route 2 (via Govindpura interior service lanes, 6.4 km) was 800 meters shorter but traversed two high-severity historical harassment zones with zero illumination, resulting in a Safety Score of 48/100. Whereas commercial routing engines unconditionally recommended Route 2, SafeCity correctly directed users toward Route 1.

C. Usability Evaluation (SUS)

Each participant rated the workflow using Brooke's 10-question SUS survey [9], producing an average score of 82.4 out of 100—well above the 70.0 benchmark for usable systems.

D. Comparative Summary

Table III summarizes the operational advantages of SafeCity over traditional commercial navigation platforms and standalone reactive panic alarms.

TABLE III. TRADITIONAL NAVIGATION / PANIC APPS VS. SAFECITY

Evaluation Dimension	Traditional Systems	SafeCity Platform
Routing Objective	Shortest Distance / Time	Safety-Distance Multi-Objective
Hazard Consideration	Traffic Flow Only	Street Harassment, Lighting, Havens
User Cognitive Load	High (Manual SOS Under Attack)	Zero (Autonomous Watchdog)
Reporting Mechanism	Formal Police Report / SMS	Anonymous Micro-Spatial Grid (~110m)
Pre-Trip Intelligence	None (Kinematic Only)	Dynamic Corridor Scoring (0-100)
Location Privacy	Commercial Ad Tracking	Quantized 3-Decimal Coordinates
Single Point of Failure	Cloud / Cellular Outage	Offline PWA Shell + Local Caching

E. Limitations of the Evaluation Plan

The experimental evaluation was conducted within the Bengaluru metropolitan region using a curated repository of verified incident reports and safe havens. Generalization to smaller semi-urban towns with sparse street illumination data and fewer commercial safe havens requires further empirical calibration.

VII. CONCLUSION AND FUTURE WORK

This paper presented SafeCity, a comprehensive community-based women protection and proactive incident reporting platform designed to eliminate the systemic shortcomings of reactive panic buttons and shortest-path navigation engines. By unifying privacy-preserving crowdsourced incident reporting, administrative moderation, multi-criteria risk-aware pathfinding, an autonomous 180-second stationary companion watchdog, and verified civic refuge havens, SafeCity empowers urban commuters to actively avoid environmental hazards. Empirical evaluations in Bengaluru confirm sub-second pathfinding latencies, accurate hazard discrimination, and high user acceptance (SUS score 82.4/100). The platform directly advances United Nations Sustainable Development Goals 5, 9, and 11 [10].

Future work will explore four key extensions: (i) lightweight on-device acoustic machine learning models to detect vocal distress or screams in real time; (ii) ad-hoc peer-to-peer Bluetooth mesh networking to broadcast emergency beacons during total cellular blackouts; (iii) zero-knowledge proof cryptography to verify community credentials without revealing citizen identities; and (iv) automated integration with municipal smart-city lighting telemetry to continuously update real-time road illumination surfaces.

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